Tharusha Sihan Fonseka

Biomedical Engineering Undergraduate, University of Moratuwa

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Education

University of Moratuwa — Moratuwa, Sri Lanka
B.Sc. Engineering (Hons.) in Biomedical Engineering; ***CGPA: 3.94/4.00*** — *Jun 2021 - Present*
Dean's List: *Semesters 1-7*

- *Robotics, Autonomous Systems, Electronic Control Systems, Advanced Electronic Control Systems.*
- *Electronic Instrumentation, Embedded Systems, Medical Electronics and Instrumentation, Electronic Circuit Design.*
- *Digital Signal Processing, Bio Signal Processing, Medical Imaging, Data Structures and Algorithms.*

Minor in Mathematics

- *Numerical Methods, Linear Algebra, Number Theory and Cryptography, Advanced Differential Equations, Linear Models and Multivariate Statistics, Calculus.*

Ananda College — Colombo 10, Sri Lanka
GCE Advanced Level - Physical Science Stream; Z-Score 3.1996; ***Island Rank -1st*** — *Jan 2017 - Aug 2019*

- *Combined Mathematics, Physics, Chemistry*

Online courses

Computer Vision (Kaggle) — *May 2025*
Intro to Deep Learning (Kaggle) — *May 2025*
ROS2 for Beginners (Udemy) — *Apr 2025*
Robotics: Aerial Robotics (University of Pennsylvania, Coursera) — *Oct 2023*
Machine Learning Specialization (DeepLearning.AI, Coursera) — *Apr 2023*

Experience

Visiting Instructor - University of Moratuwa — *Feb 2025 - Present*

- Conducting laboratory practicals and tutorial classes of EN2533- Robot design and competition, EN3563 - Robotics, and En4594 Autonomous systems modules.
- Conducted electronic laboratory practicals of EN1094 - Laboratory practice module.

Consultant Electronic Engineer - Pekoe Pte Ltd. — *Jul 2025 - Present*

- Pekoe Pte Ltd. is a research group aiming to introduce AI solutions to the tea industry.
- Responsible for designing instruments and methods to measure the physical properties of tea leaves for grading.

Research Affiliate - University of Sydney — *Dec 2023 - May 2024*

- Simulated the stretching of epidermal sensors with different body motions to analyze the stress and strain.
- Designed a wearable device to capture and transmit real-time resistance data from an epidermal stretch sensor.
- Setup the hardware for simultaneous data capturing from two radar devices in different frequency bands to create a synthetic band using machine learning.

Projects

Humanoid Head For Robot Receptionist Capable of Natural Human Interaction — *Jun 2024 - May 2025*

- Project team lead.
- The robot head is capable of bio mimicking facial expressions, active eye gazing, human re-identification, and natural conversation abilities.
- I worked on hardware design, implementation, and ROS2-based controlling.

Phoenix Cube Satellite Project — *Ongoing*

- Stage 1, launching a high-altitude balloon into the stratosphere.
- I am leading a team of twenty undergraduates.

Re-implementing research paper on Bio Signal Processing — *2024*

- Re-implemented the paper "Classification of ECG heartbeats using nonlinear decomposition methods and support vector machine".

Maze Solving Micromouse Robot(Robofest 2023, 2024) — *2023, 2024*

- Fully Autonomous maze solving robot, which is tasked to reach the destination point in the shortest time possible.
- I contributed to the sensor fusion, control system, and implementation of the maze-solving algorithm.

Scorpion - Battle Robot (EXMO 2023) *Jul 2023*

- Designed a battle robot to demonstrate in EXMO 2023.
- I was involved in designing and implementing the mechanical and structural design.

Smart Switch for Automating Conventional Home Appliances *Jun 2023*

- Designed a smart switch to automate conventional home appliances, which can be easily integrated into an existing domestic installation.
- Can be controlled remotely through internet or manually through capacitive touch buttons.
- An individual project.

Analog Implementation of a Five Band Audio Equalizer *Feb 2023*

- Five band audio equalizer using analog 4^{th} order Sallen Key filters and a differential amplifier.

Awards and Scholarships

Winners | Electronic Design Competition *Sep 2023*
IEEE Sri Lanka section

1^{st} Runners-up | SPARK Challenge 2023 *Jun 2023*
Innovation Competition for Sustainable Development - Electronic Club of University of Moratuwa

Dialog Merit Scholarship Award *2023*
For outstanding performance in GCE A/L Examination

S.A. Wijethilake Memorial Challenge Cup | Ananda College *2020*
Awarded for the Best Academic Performance of the College at the G.C.E. A/L Examination 2019

"Anada Pradeepa" Award | Ananda College *2020*
Awarded to the College subject toppers based on the G.C.E. A/L Examination 2019

Silver medalist | Sri Lankan Physics Olympiad *2019*
Awarded by Institute of Physics Sri Lanka

Silver medalist | Sri Lankan Astronomy and Astrophysics Olympiad *2018*
Awarded by Institute of Physics Sri Lanka

Leadership and Volunteering

Project Manager - SEDS-Mora - University of Moratuwa *Apr 2024 - Apr 2025*

Old Anandian Engineers' Guild - Students' Chapter
Secretary *Mar 2024 - Apr 2025*
Vice President *Mar 2023 - Mar 2024*
Deputy Director- School Affairs *Jan 2022 - Mar 2023*

Event Team Lead - Brainstorm 2023 - IEEE EMBS, University of Moratuwa *Aug 2022 - May 2023*

Field Representative - Department of ENTC - University of Moratuwa *May 2022 - Feb 2023*

Event Co-chair - TourEye - Roteract Club - University of Moratuwa *Sept 2021*

Chief Organizer - Anandian Astronomical Association - Ananda College *Nov 2017 - Nov 2018*

Skills Summary

Languages: English, Sinhala (Native Proficiency)
Programming Languages: Python, Matlab, C++
Frameworks: ROS2
Software: PCB designing (Altium), CAD designing (Solidworks), Circuit simulation (Proteus, LtSpice)
Soft Skills: Leadership, Project Management, Team working

References

Dr. Peshala Jayasekara
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